Denys Tsinyk

(267) 982-1127 | denystsinyk@gmail.com | github.com/denystsinyk | linkedin.com/in/denystsinyk

Education

University of Pittsburgh December 2027
Bachelors of Science in Computer Science *Pittsburgh, PA*

- GPA: 3.8/4.0
- Courses: Data Structures & Algorithms, Discrete Structures, Assembly Language & Computer Organization, Web Development, Foundations of Machine Learning, System Software, Cloud Computing, Software Engineering

Experience

Character.AI May 2026 - August 2026
Software Engineer Intern *Redwood City, CA*

- Architected a production safety pipeline for user-generated content, reducing publish latency from 5 minutes to 20 seconds through Redis-backed caching, content hashing, and OpenAI moderation integration.
- Built a multimodal entity resolution system for 250M AI characters, identifying copyrighted characters and public figures using image/text embeddings, FAISS indexing, and a Spanner-backed registry.

National Football League (NFL) June 2025 - August 2025
Software Engineer Intern *Mount Laurel, NJ*

- Built the NFL's first automated security configuration assessment pipeline, detecting misconfigurations across 3,000+ servers through Qualys Cloud Agent deployment.
- Reduced Center for Internet Security (CIS) benchmark validation time by ∼57% by automating Qualys result parsing, compliance validation, and report generation in Python.
- Developed a Python analysis tool for 5,000+ firewall rules, automating source-destination flow mapping and reducing incident response time by 30%.

MedPath January 2025 – May 2025
Software Engineer Intern *Pittsburgh, PA*

- Built a full-stack career planning platform that helped students evaluate medical career pathways through personalized assessments, course planning, and real-time progress tracking.
- Designed scalable backend services and relational data models supporting secure OAuth authentication, interactive learning modules, and student dashboards using Node.js and PostgreSQL.
- Developed responsive frontend experiences in React, enabling students to navigate assessments, educational content, and progress analytics through a unified interface.

University of Pittsburgh August 2025 - Present
Undergraduate Teaching Assistant *Pittsburgh, PA*

- Lead weekly recitations for 30+ students, teaching Data Structures & Algorithms through live coding, code reviews, and exercises covering trees, graphs, hashing, dynamic programming, and algorithm analysis.

Projects

FlyWheel

- Built a content generation platform capable of matching writing styles with 85% voice similarity.
- Engineered an MCP-powered research pipeline integrating Claude and Perplexity APIs for planning, fact-checking, citation retrieval, and autonomous content generation.
- Designed a pgvector-powered retrieval architecture with normalized PostgreSQL schemas, reducing retrieval latency by 40% for RAG-based generation.

Pitt2PIT

- Built a ride-sharing platform connecting students traveling to overlapping destinations, reducing projected travel costs by up to 73%.
- Designed a ride-matching algorithm that optimized departure windows, destination overlap, vehicle capacity, and detour distance to maximize pairing quality.
- Implemented secure authentication, ride coordination, and in-app messaging using React, FastAPI, and PostgreSQL.

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, C, C++, HTML, CSS, Go, SQL
Libraries/Frameworks: Node.js, Next.js, React, Flask, pandas, FastAPi, OpenAPI
Developer Tools: Git, Docker, Kubernetes, Spanner, Redis, Prometheus